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Association of frailty and cognitive function disorders in patients with COVID-19: a protocol of systematic review and meta-analysis

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-056190
Article Type:	Protocol
Date Submitted by the Author:	07-Aug-2021
Complete List of Authors:	jiao, bo; West China Hospital of Medicine Chen, Mingyuan; Sichuan University West China Second University Hospital, Department of Pathology Luo, Xi; Sichuan University West China Hospital, Department of Anesthesiology Fan, Minmin; Sichuan University, Department of Anesthesiology Chen, Chan; Sichuan University West China Hospital, Liu, Jin; Sichuan University West China Hospital, Department of Anesthesiology
Keywords:	COVID-19, GERIATRIC MEDICINE, MENTAL HEALTH, Delirium & cognitive disorders < PSYCHIATRY, Old age psychiatry < PSYCHIATRY

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Association of frailty and cognitive function disorders in patients with COVID-19: a protocol of systematic review and meta-analysis

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Abstract

Introduction

COVID-19 infection has become a worldwide urgent public health concern. COVID-19 is primarily a respiratory disease but up to two thirds of hospitalised patients show evidence of nervous system damage and increase the risk of frailty. We aim to investigate the association between frailty and cognitive function disorders in patients with COVID-19 by this systematic review and meta-analysis.

Methods and Analysis

This meta-analysis has been registered in the International Prospective Register of Systematic Reviews (PROSPERO). We will search PubMed, Embase, Chinese Biological Medical Database (CBM), China National Knowledge Infrastructure (CNKI), Wanfang Database, the Cochrane Central Register of Controlled Trials (CENTRAL). We also will search reference lists of selected articles. We used no language restrictions. We will use a fixed or random-effects model to calculate the odds ratio (OR) and 95% confidence interval (CI) for pooled data. We will assessed heterogeneity using Cochrane's Q and I^2 tests. The primary outcome is he rate of cognitive disorders related to frailty in old patients with COVID-19.

Ethics and Dissemination

Ethical approval is not essential for the reason that the data will be extracted from the previously published studies. Lastly, the results of this meta-analysis will be published in peer-reviewed journal.

PROSPERO registration number: CRD42021257148

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Strength and Limitations of this review

1.This is the first systematic review to investigate the association of frailty and cognitive function disorders in patients with COVID-19.

2.This meta-analysis will provide a evidence to examine the association between frailty and cognitive function disorders in old patients with COVID-19. And to help doctors to adopt some measures to prevent the effect of frailty on cognition.

3. The number of original studies is limited, thus, the strength of evidence needs to be examined cautiously.

Key Words: COVID-19, Frailty, Cognitive impairment, Elderly persons

Introduction

With the novel coronavirus pneumonia recently broke out in India once again, this epidemic still needs to be paid great attention because it seriously threatens health of people in the world[1, 2]. Under the influence of COVID-19, the function of various organs or systems will be damaged, such as the central nervous system (CNS). The potential mechanism might be that more inflammatory cytokines enter into the brain and cause the CNS injure[3-5]. Therefore, the people who have novel coronavirus infection may emerge with neurologic and mental disorders, especially the elderly[6, 7]. The main manifestations are as follows: cognitive deficits, attention deficits, circadian rhythm disorders, mood disorders, and changes in psychomotor function[8]. Cognitive function disorder is a serious illness that could affect various aspects in patients, such as mortality, et.al. Moreover, with the cognitive function decline, the mortality of patients will be increasing, especially in the population that age is

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between 65 and 79 years old[9]. Frailty is also a healthy burden, which can increase the incidence of other diseases and medical expenses, and it is an age-related clinical state characterized by deterioration in the physiological capacity of organs and systems[10]. The risk of frailty will be greatly increased with age[11]. Therefore, the elderly with frailty, suffering from novel coronavirus pneumonia, are more likely to have cognitive impairment. And the potential mechanistic links between them may be inflammation and oxidative stress[12]. However, another existing study had found that the COVID-19 could accelerate the aging process and increase frailty, but the cognitive function did not decline.[13]. And then, the association between frailty and cognitive function disorders in patients with COVID-19 needs more reliable evidence to be proved.

Whether reducing the rate of frailty by enhancing the state of body can improve the old patients' cognitive function is significantly important for clinical work. Therefore, it is necessary to find out the association among them in these patients with COVID-19. As far as we know, no systematic review and meta-analysis have estimated the association between frailty and cognitive disorders in patients with COVID-19. Hence, we will qualitatively and quantitatively investigate the effect of the frailty on cognitive disorders in old patients with novel coronavirus pneumonia in this systematic review and meta-analysis.

Methods

This protocol of meta-analysis was performed abide by the Preferred Reporting Items for Systematic Reviews and Meta-analyses Protocols (the PRISMA-P)[14]. The PROSPERO registration number is CRD42021257148. All analyses were based on

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previously published studies, thus no ethical approval and patient consent are required.

Patient and public involvement

No patient involved.

Literature search

We will search the database, including PubMed, Embase, Chinese Biological Medical Database (CBM), China National Knowledge Infrastructure (CNKI), Wanfang Database, the Cochrane Central Register of Controlled Trials (CENTRAL). We will also search reference lists of selected articles. We used no language restrictions. All studies that investigate the association of frailty and cognitive function in patients with COVID-19 will be enrolled in this systematic review, including prospective and retrospective comparative cohort studies, cross-sectional researches, and observational researches. 'COVID-19 Virus Disease' or 'COVID-19' or 'SARS-CoV-2 and 'frailty' and 'cognition disorders' are the search terms, and the full search approach of PubMed is presented in Table 1.

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Table 1 search strategy of PubMed

Search	Query
#1	“COVID-19”[MeSH Terms]
#2	"covid 19"[MeSH Terms] OR "covid 19"[All Fields] OR "covid 19 virus disease"[All Fields] OR "covid 19"[MeSH Terms] OR "covid 19"[All Fields] OR "disease covid 19 virus"[All Fields] OR "covid 19"[MeSH Terms] OR "covid 19"[All Fields] OR "covid 19 virus infection"[All Fields] OR "covid 19"[MeSH Terms] OR "covid 19"[All Fields] OR "covid 19"[MeSH Terms] OR "covid 19"[All Fields] OR "2019 ncov infection"[All Fields] OR "covid 19"[MeSH Terms] OR "covid 19"[All Fields] OR "infection 2019 ncov"[All Fields] OR "covid 19"[MeSH Terms] OR "covid 19"[All Fields] OR "coronavirus disease 19"[All Fields] OR "covid 19"[MeSH Terms] OR "covid 19"[All Fields] OR "2019 novel coronavirus disease"[All Fields] OR "covid 19"[MeSH Terms] OR "covid 19"[All Fields] OR "2019 novel coronavirus infection"[All Fields] OR "covid 19"[MeSH Terms] OR "covid 19"[All Fields] OR "2019 ncov disease"[All Fields] OR "covid 19"[MeSH Terms] OR "covid 19"[All Fields] OR "disease 2019 ncov"[All Fields] OR "covid 19"[MeSH Terms] OR "covid 19"[All Fields] OR "covid19"[All Fields] OR "covid 19"[MeSH Terms] OR "covid 19"[All Fields] OR "coronavirus disease 2019"[All Fields] OR "covid 19"[MeSH Terms] OR "covid 19"[All Fields] OR "sars coronavirus 2 infection"[All Fields] OR "covid 19"[MeSH Terms] OR "covid 19"[All Fields] OR "sars cov 2 infection"[All Fields] OR "covid 19"[MeSH Terms] OR "covid 19"[All Fields] OR "covid 19 pandemic"[All Fields] OR "covid 19"[MeSH Terms] OR "covid 19"[All Fields] OR "pandemic covid 19"[All Fields]
#3	#1 or #2
#4	"frailty"[MeSH Terms]
#5	"frailty"[MeSH Terms] OR "frailty"[All Fields] OR "frailties"[All Fields] OR ("frail"[All Fields] OR "frails"[All Fields] OR "frailty"[MeSH Terms] OR "frailty"[All Fields] OR "frailness"[All Fields]) OR ("frailty"[MeSH Terms] OR "frailty"[All Fields] OR ("frailty"[All Fields] AND "syndrome"[All Fields]) OR "frailty syndrome"[All Fields]) OR ("frailty"[MeSH Terms] OR "frailty"[All Fields] OR "debility"[All Fields]) OR ("frailty"[MeSH Terms] OR "frailty"[All Fields] OR "debilities"[All Fields])
#6	#4 or #5
#7	"cognition disorders"[MeSH Terms]
#8	"cognition disorders"[MeSH Terms] OR ("cognition"[All Fields] AND "disorders"[All Fields]) OR "cognition disorders"[All Fields] OR ("disorder"[All Fields] AND "cognition"[All Fields]) OR "disorder cognition"[All Fields] OR ("cognition disorders"[MeSH Terms] OR ("cognition"[All Fields] AND "disorders"[All Fields]) OR "cognition disorders"[All Fields] OR ("disorders"[All Fields] AND "cognition"[All Fields]) OR "disorders cognition"[All Fields]) OR ("cognition disorders"[MeSH Terms] OR ("cognition"[All Fields] AND "disorders"[All Fields]) OR "cognition disorders"[All Fields] OR "overinclusion"[All Fields] OR "overinclusive"[All Fields] OR "overinclusiveness"[All Fields])
#9	#7 or #8
#10	#3 and #6 and #9

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Eligibility criteria

Type of study design

All relevant studies, including prospective and retrospective comparative cohort studies, cross-sectional researches, and observational researches, will be contained in this systematic review. Because original trials are not abundant, we will acquire as many related studies as possible by improving the search strategy sensitivity.

Participants

The patients with COVID-19 who have cognitive impairment will be involved. The subjects who are used to investigate the relationship between frailty and cognition disorders, and the confounding factors which will affect the rate of cognitive function disorders must be emended. We will not restrict the race because of the limitations in the number of original studies.

Exposures

The main exposure factor is whether there is frailty performance. And the score of frailty will be assessed by the relevant scale, not only the 'FRAIL' scale.

Outcomes

The primary outcome is the rate of cognitive disorders related to frailty in old patients with COVID-19. The secondary outcomes are the rate of mortality, length of hospitalization.

Study selection and data collection

The studies will be selected by two reviewers independently. When there are some differences between the reviewers, further discussions will be held with a senior

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reviewer. After removing the duplicate items, we will evaluate the preliminary quality of the study based on the title and abstract. Then, we will make the final evaluation by reading the full text. The selection flow gram is presented in Fig.1. A standard collection form will be used and the data will be extracted independently by two reviewers. Any discrepancies will be resolved by another senior reviewer. Data extracted included first author, date of publication, sample size, patient characteristics, type of study, number of patients with frailty, grade of frailty, number of patients with cognition disorders (recorded separately in two groups), number of patients without cognition disorders (recorded separately in two groups), number of deaths, rate of mortality, length of hospital stay. For the missing data, we will contact the first author to acquire it.

Risk-of-bias assessment and Quality assessed

The quality and bias of studies involved will be assessed with two tools. The Newcastle-Ottawa Scale (NOS) will be used to evaluate the bias of the non-randomized studies according to the following fields: objects' selection, study comparability, and outcome measurement[15]. The decisions could be determined as 'Low', or 'High' risk of bias, or 'Some concerns' along with the tool of evaluating the risk of bias. Two reviewers will assess the risk of bias of each study included independently. The tool, named Grading of Recommendations, Assessment, Development, and Evaluation (GRADE) criteria, will be applied to evaluate the strength of evidence[16].

Data synthesis and statistical analysis

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Data synthesis and meta-analysis will be performed with the Review Manager 5.4 software (The Cochrane library, Oxford, England). The odds ratio (OR) and mean difference (MD) with 95% confidence interval (CI) were employed to analyze the categories and continuous data. Heterogeneity across studies was assessed with chi-square test and I^2 test, and $I^2 > 50\%$ or $P < 0.10$ was considered as significantly heterogenous. The random-effect model was adopted if the heterogeneity existed among the studies, whereas the fixed-effect model was applied if no significant heterogeneity was detected. [17].

Subgroup analysis, sensitivity analysis, and publication bias assessment

When there is a high degree of heterogeneity, the subgroup analysis will be performed relying on the level of frailty. If each group covers two studies at least, the subgroup analysis will be performed, otherwise, it does not be executed. Sensitivity analysis will be performed by removing the studies which are low-quality, if the final pooled results are stable after removing the low-quality studies, the meta-analysis will be deemed as reliable[18]. The publication bias will be investigated with funnel plot and Egger's test[19,20].

Discussion

According to our knowledge, this meta-analysis is the first one to investigate the association between frailty and cognitive function disorders. It is meaningful for old patients with COVID-19 to find out the effect of frailty on the incidence of cognitive disorders.

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The incidence of frailty will increase significantly with age. Frailty is a reduction in the ability to make a response to stress, and the elderly with this condition are more vulnerable. And cognitive impairment is a serious threat to healthy living. In recent studies, it has been verified that age is the most important independent risk factor for cognitive disorders[21]. Due to the influence of COVID-19, serious inflammation is existing in these patients, and inflammatory factors may increase the risk of nerve injure. Therefore, it is important to investigate the association between them in patients with novel coronavirus pneumonia.

There are several limitations in our meta-analysis. First, the number of original studies is limited, thus, the strength of evidence needs to be examined in the future. Secondly, the tools to assess the cognitive impairment are various, which can increase the heterogeneity of studies included. Finally, due to the limitation of original studies, we will not make the other subgroup analysis except the level of frailty, and then the heterogeneity may not be explained fully if a high degree of heterogeneity exists in this meta-analysis.

Conclusion

This manuscript was a protocol of meta-analysis, and the details of association between frailty and cognitive disorders in patients with COVID-19 had been described. The specific methods of performing meta-analysis was provided in the paper. It has been registered in PROSPERO and the final results would be published when it accepted a peer review.

Acknowledgements

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Not applicable.

Contributors

BJ and XL designed this review and made the search strategy. BJ and MYC will do the literature search, data collection. BJ and MMF will perform the data synthesis and analysis, and BJ was major contributor to write the protocol draft. JL and CC revised the manuscript critically. All authors have read and approved the final version.

Funding

This review did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Patient consent

Not applicable.

Competing interests

No authors declare competing interests.

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Figure legends

Fig.1 The flow diagram of literature search and selection

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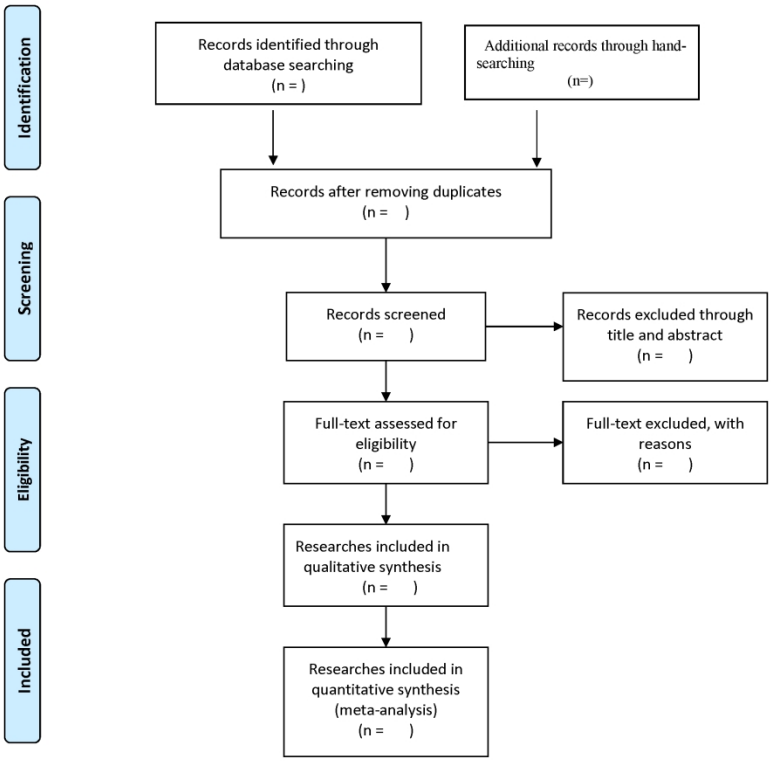


Fig.1 The flow diagram of literature search and selection

215x279mm (300 x 300 DPI)

PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol*

Section and topic	Item No	Checklist item
ADMINISTRATIVE INFORMATION		
Title:		
Identification	1a	Identify the report as a protocol of a systematic review (yes)
Update	1b	If the protocol is for an update of a previous systematic review, identify as such（no）
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number(yes, CRD42021257148)
Authors:		
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author(yes)
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review(yes)
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments(yes)
Support:		
Sources	5a	Indicate sources of financial or other support for the review(yes)
Sponsor	5b	Provide name for the review funder and/or sponsor(yes)
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol(yes)
INTRODUCTION		
Rationale	6	Describe the rationale for the review in the context of what is already known(yes)
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO) (yes)
METHODS		
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review (yes)
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage (yes)
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated (yes)
Study records:		
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review (yes)

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Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis) (yes)
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators (yes)
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications (yes)
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale (yes)
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis (yes)
Data synthesis	15a	Describe criteria under which study data will be quantitatively synthesised (yes)
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I^2, Kendall's τ) (yes)
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression) (yes)
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned (yes)
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies) (yes)
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE) (yes)

*** It is strongly recommended that this checklist be read in conjunction with the PRISMA-P Explanation and Elaboration (cite when available) for important clarification on the items. Amendments to a review protocol should be tracked and dated. The copyright for PRISMA-P (including checklist) is held by the PRISMA-P Group and is distributed under a Creative Commons Attribution Licence 4.0.**

From: Shamseer L, Moher D, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart L, PRISMA-P Group. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation. BMJ. 2015 Jan 2;349(jan02 1):g7647.

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Association of frailty and cognitive function disorders in old patients with COVID-19: a protocol of systematic review and meta-analysis

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-056190.R1
Article Type:	Protocol
Date Submitted by the Author:	15-Dec-2021
Complete List of Authors:	jiao, bo; West China Hospital of Medicine Chen, Mingyuan; Sichuan University West China Second University Hospital, Department of Pathology Fan, Minmin; Sichuan University, Department of Anesthesiology Luo, Xi; Sichuan University West China Hospital, Department of Anesthesiology Chen, Chan; Sichuan University West China Hospital, Liu, Jin; Sichuan University West China Hospital, Department of Anesthesiology
Primary Subject Heading:	Geriatric medicine
Secondary Subject Heading:	Global health, Infectious diseases, Mental health
Keywords:	COVID-19, GERIATRIC MEDICINE, MENTAL HEALTH, Delirium & cognitive disorders < PSYCHIATRY, Old age psychiatry < PSYCHIATRY

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Association of frailty and cognitive function disorders in old patients with COVID-19: a protocol of systematic review and meta-analysis

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Abstract

Introduction

COVID-19 infections have become an urgent worldwide public health concern. Although it is primarily a respiratory disease, up to two thirds of hospitalised COVID-19 patients exhibit nervous system damage and an increased risk of frailty. In this study,we aim to investigate the relationship between frailty and cognitive function disorders in patients with COVID-19 with a systematic review and meta-analysis approach.

Methods and analysis

This meta-analysis has been registered by the International Prospective Register of Systematic Reviews (PROSPERO). We will search for relevant studies from PubMed, Embase, Chinese Biological Medical Database (CBM), China National Knowledge Infrastructure (CNKI), Wanfang Database, the Cochrane Central Register of Controlled Trials (CENTRAL) databases, from their inception until 05 July 2021. We will also search reference lists of selected articles for additional studies. Our search strategy will have no language restrictions. We will employ a fixed or random-effects model to calculate odds ratio (OR) and 95% confidence intervals (CI) for pooled data, and assess heterogeneity using Cochrane's Q and I^2 tests. The primary outcome will be the rate of cognitive disorders related to frailty in old patients with COVID-19.

Ethics and dissemination

Ethical approval is not essential since data will be extracted from previously published studies. The results of this meta-analysis will be published in a peer-reviewed journal.

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PROSPERO registration number: CRD42021257148

Strengths and limitations of this review

1. This is the first systematic review seeking to elucidate the relationship between frailty and cognitive function disorders in patients with COVID-19.

2. The findings of this study will provide relevant insights to guide doctors to adopt some measures to prevent the effect of frailty on cognition.

3. The number of original studies is limited, thus the strength of evidence needs to be examined cautiously.

Key Words: Cognitive impairment; COVID-19; elderly persons; frailty.

Introduction

The recent wave of infections by the novel coronavirus in India calls for urgent and numerous attention to be paid to this epidemic owing to the serious threat on human health worldwide.[1 2] COVID-19 infection damages the function of various organs or systems, such as the central nervous system (CNS), possibly through infiltration of inflammatory cytokines into the brain that cause injury to the CNS.[3-5] Consequently, people who contract novel coronavirus infections, especially the elderly, may experience neurologic and mental disorders,[6 7] with the following as main manifestations: cognitive and attention deficits, circadian rhythm and mood disorders, as well as changes in psychomotor function.[8] Cognitive function disorder is a serious illness that negatively impacts patients' lives, including causing deaths. Moreover, a decline in cognitive function has been associated with increased patient mortality, especially in people aged between 65 and 79 years.[9]

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Frailty, a health burden that exacerbates incidence of other diseases and medical expenses, is an age-related clinical state characterized by deterioration in the physiological capacity of organs and systems.[10] Previous studies have shown that the risk of frailty greatly increases with age,[11] suggesting that elderly people with frailties, who suffer from novel coronavirus pneumonia, are likely to be at a greater risk of cognitive impairment. Research evidence has suggested that this phenomenon may potentially be attributed to inflammation and oxidative stress.[12] However, findings from another study suggested that COVID-19 could accelerate the aging process and increase frailty, albeit with no decline in cognitive function.[13] Therefore, there is need to elucidate the relationship between frailty and cognitive function disorders in patients with COVID-19.

Particularly, knowledge on whether reducing the rate of frailties by enhancing the state of the body in elderly COVID-19 patients can improve their cognitive function is imperative to effective clinical therapies. As far as we know, no systematic review and meta-analysis has evaluated the relationship between frailty and cognitive disorders in patients with COVID-19, necessitating further explorations. In the present study, we aim to qualitatively and quantitatively investigate the effect of frailty on cognitive disorders in elderly patients with novel coronavirus pneumonia using a systematic review and meta-analysis.

Methods

This meta-analysis will be performed in accordance to the guidelines by the Preferred Reporting Items for Systematic Reviews and Meta-analyses Protocols (the PRISMA-

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P)[14] and is registered under PROSPERO (registration number is CRD42021257148). All analyses will be based on previously published studies, thus neither ethical approval nor patient consent will be required.

Literature search

We will search for and retrieve relevant studies across various databases, including PubMed, Embase, Chinese Biological Medical Database (CBM), China National Knowledge Infrastructure (CNKI), Wanfang Database, and the Cochrane Central Register of Controlled Trials (CENTRAL). We will also search reference lists of the selected articles to obtain additional studies. There are no language restrictions during the search process, and all studies describing the relationship between frailty and cognitive function in patients with COVID-19 will be enrolled in this systematic review. These will include prospective and retrospective comparative cohort studies, cross-sectional researches, and observational researches. The search strategy will employ the following terms: 'COVID-19 Virus Disease' or 'COVID-19' or 'SARS-CoV-2 and 'frailty' and 'cognition disorders'. A summary of the full search approach of PubMed is outlined in Table 1.

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Table 1 search strategy of PubMed

Search	Query
#1	"COVID-19"[MeSH Terms]
#2	"covid 19"[MeSH Terms] OR "covid 19"[All Fields] OR "covid 19 virus disease"[All Fields] OR "covid 19"[MeSH Terms] OR "covid 19"[All Fields] OR "disease covid 19 virus"[All Fields] OR "covid 19"[MeSH Terms] OR "covid 19"[All Fields] OR "covid 19 virus infection"[All Fields] OR "covid 19"[MeSH Terms] OR "covid 19"[All Fields] OR "covid 19"[MeSH Terms] OR "covid 19"[All Fields] OR "2019 ncov infection"[All Fields] OR "covid 19"[MeSH Terms] OR "covid 19"[All Fields] OR "infection 2019 ncov"[All Fields] OR "covid 19"[MeSH Terms] OR "covid 19"[All Fields] OR "coronavirus disease 19"[All Fields] OR "covid 19"[MeSH Terms] OR "covid 19"[All Fields] OR "2019 novel coronavirus disease"[All Fields] OR "covid 19"[MeSH Terms] OR "covid 19"[All Fields] OR "2019 novel coronavirus infection"[All Fields] OR "covid 19"[MeSH Terms] OR "covid 19"[All Fields] OR "2019 ncov disease"[All Fields] OR "covid 19"[MeSH Terms] OR "covid 19"[All Fields] OR "disease 2019 ncov"[All Fields] OR "covid 19"[MeSH Terms] OR "covid 19"[All Fields] OR "covid19"[All Fields] OR "covid 19"[MeSH Terms] OR "covid 19"[All Fields] OR "coronavirus disease 2019"[All Fields] OR "covid 19"[MeSH Terms] OR "covid 19"[All Fields] OR "sars coronavirus 2 infection"[All Fields] OR "covid 19"[MeSH Terms] OR "covid 19"[All Fields] OR "sars cov 2 infection"[All Fields] OR "covid 19"[MeSH Terms] OR "covid 19"[All Fields] OR "covid 19 pandemic"[All Fields] OR "covid 19"[MeSH Terms] OR "covid 19"[All Fields] OR "pandemic covid 19"[All Fields]
#3	#1 or #2
#4	"frailty"[MeSH Terms]
#5	"frailty"[MeSH Terms] OR "frailty"[All Fields] OR "frailties"[All Fields] OR ("frail"[All Fields] OR "frails"[All Fields] OR "frailty"[MeSH Terms] OR "frailty"[All Fields] OR "frailness"[All Fields]) OR ("frailty"[MeSH Terms] OR "frailty"[All Fields] OR ("frailty"[All Fields] AND "syndrome"[All Fields]) OR "frailty syndrome"[All Fields]) OR ("frailty"[MeSH Terms] OR "frailty"[All Fields] OR "debility"[All Fields]) OR ("frailty"[MeSH Terms] OR "frailty"[All Fields] OR "debilities"[All Fields])
#6	#4 or #5
#7	"cognition disorders"[MeSH Terms]
#8	"cognition disorders"[MeSH Terms] OR ("cognition"[All Fields] AND "disorders"[All Fields]) OR "cognition disorders"[All Fields] OR ("disorder"[All Fields] AND "cognition"[All Fields]) OR "disorder cognition"[All Fields] OR ("cognition disorders"[MeSH Terms] OR ("cognition"[All Fields] AND "disorders"[All Fields]) OR "cognition disorders"[All Fields] OR ("disorders"[All Fields] AND "cognition"[All Fields]) OR "disorders cognition"[All Fields]) OR ("cognition disorders"[MeSH Terms] OR ("cognition"[All Fields] AND "disorders"[All Fields]) OR "cognition disorders"[All Fields] OR "overinclusion"[All Fields] OR "overinclusive"[All Fields] OR "overinclusiveness"[All Fields])
#9	#7 or #8
#10	#3 and #6 and #9

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Eligibility criteria

Study design

All relevant studies, including prospective and retrospective comparative cohort studies, cross-sectional researches, and observational researches, will be included in this systematic review. Due to scarcity of original trials on COVID-19, we will acquire as many related studies as possible by improving the sensitivity of the search strategy.

Participants

Elderly COVID-19 patients ($\geqslant$ 65 years old), with cognitive impairment, will be considered. The subjects who are used to investigate the relationship between frailty and cognition disorders, and the confounding factors that affect the rate of cognitive function disorders will be emended. There will be no restriction on the race due to expected limitations in the number of original studies.

Exposures

The main exposure factor will be whether presence of frailty. Therefore, we will assess the frailty scores using a relevant scale, and not just the 'FRAIL' scale only.

Outcomes

The primary outcome will be the rate of cognitive disorders in elderly COVID-19 patients with frailty. Cognitive disorder will be assessed using the Brief Assessment of Impaired Cognition (BASIC) tool, which focuses on the self and informant reports,

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supermarket fluency, and category cued memory test.[15] Secondary outcomes will comprise the rate of mortality, and length of hospitalization.

Study selection and data collection

Relevant studies will be independently searched and selected by two reviewers, and any differences between them resolved via further discussions with a senior reviewer. After removing the duplicate items, we will evaluate the preliminary quality of the studies based on the title and abstract. Then, we will make the final evaluation by reading the full text. A schematic representation of the selection procedure is presented in Fig.1. A standard collection form will be used for data retrieval, which will also be independently done by the two reviewers. Any discrepancies between them will be resolved through discussion with the aforementioned senior reviewer. Data to be extracted will include, name of the first author, date of publication, sample size, patient characteristics, type of study, number of patients with frailty, grade of frailty, number of patients with cognition disorders (recorded separately in frailty and non-frailty group), number of patients without cognition disorders (recorded separately in frailty and non-frailty group), number of deaths, rate of mortality, and length of hospital stay. In cases where there is missing data, we will attempt to contact the corresponding author to acquire it.

Determination of risk-of-bias and assessment of study quality

Quality and bias of the enrolled studies will be assessed using relevant tools. Specifically, the Newcastle-Ottawa Scale (NOS) will be used to evaluate bias in non-randomized studies targeting the following aspects: objects' selection, study

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comparability, and outcome measurement.[16] The decisions will be determined as 'Low', or 'High' risk of bias, or 'Some concerns' along with the tool of evaluating the risk of bias. Risk of bias in each included study will be independently assessed by two reviewers. In addition, risk of bias and quality assessment will also be determined using the following measures: Inconsistency, Indirectness, Imprecision, Sensitivity, Specificity and Pooled C-statistic tool.[17] Furthermore, we will apply the Grading of Recommendations, Assessment, Development, and Evaluation (GRADE) criteria to evaluate strength of the evidence.[18]

Data synthesis and statistical analysis

Data synthesis and meta-analysis will be performed using the Review Manager 5.4 software (The Cochrane library, Oxford, England). Briefly, we will calculate odds ratio (OR) and mean differences (MD) with 95% confidence intervals (CI) for categorical and continuous data. On the other hand, we will calculate heterogeneity across the included studies using the chi-square and I^2 tests, with $I^2 > 50\%$ or $P < 0.10$ to be considered significantly heterogenous. We will adopt the random-effects model where heterogeneity exists among the studies, otherwise, the fixed-effect model will be applied.[19]

Subgroup and sensitivity analyses, and assessment of publication bias

Where there is a high degree of heterogeneity, we will perform subgroup analysis based on the level of frailty and age group. Subgroup analysis will only be performed in groups that have at least each group two studies. On the other hand, sensitivity analysis will be performed by excluding low-quality studies, and if the final pooled results are

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stable after this exclusion, then the meta-analysis will be deemed reliable.[20] Publication bias will be investigated by generating funnel plots and the Egger's test as previously described.[21 22]

Discussion

As far as we know, this is the first meta-analysis to investigate the association between frailty and cognitive function disorders. The findings of the study will provide relevant insights into the effect of frailty on the incidence of cognitive disorders in elderly COVID-19 patients. Frailty incidence significantly increases with increase in age. Frailty, which refers to a reduction in the ability to respond to stress, is more common in elderly. And cognitive impairment is a serious threat to healthy living. Previous studies have shown that age is the most important independent risk factor for occurrence of cognitive disorders.[23] Notably, COVID-19 causes serious inflammation in these patients, while inflammatory factors may increase the risk of nerve injure. Therefore, it is important to understand the relationship between frailty and cognitive function disorders in patients with novel coronavirus pneumonia.

We anticipate several limitations in the present meta-analysis. Firstly, the number of original studies is limited, which might affect the strength of the obtained evidence. Secondly, the non-randomized studies will be included, which might affect the interpretation of final pooled results. Finally, considering the limitation of original studies, we may not perform any other subgroup analysis apart from the level of frailty and age group, thus the heterogeneity may not be fully explained if a high degree of heterogeneity exists in this meta-analysis.

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Conclusion

This meta-analysis is expected to reveal the relationship between frailty and cognitive disorders in elderly COVID-19 patients.

Patient and Public Involvement

No patients involved.

Ethics and dissemination

Ethical approval is not essential since all data will be extracted from previously published studies. Results of this meta-analysis will be published in a peer-reviewed journal.

Acknowledgements

Not applicable.

Contributors

BJ and XL designed this review and made the search strategy. BJ and MYC will do the literature search, data collection. BJ and MMF will perform the data synthesis and analysis, and BJ was major contributor to write the protocol draft. BJ and MYC have contributions to this revised manuscript. JL and CC revised the manuscript critically. All authors have read and approved the final version.

Funding

This review did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Patient consent

Not applicable.

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Competing interests

No authors declare competing interests.

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Figure legends

Fig.1 The flow diagram of literature search and selection

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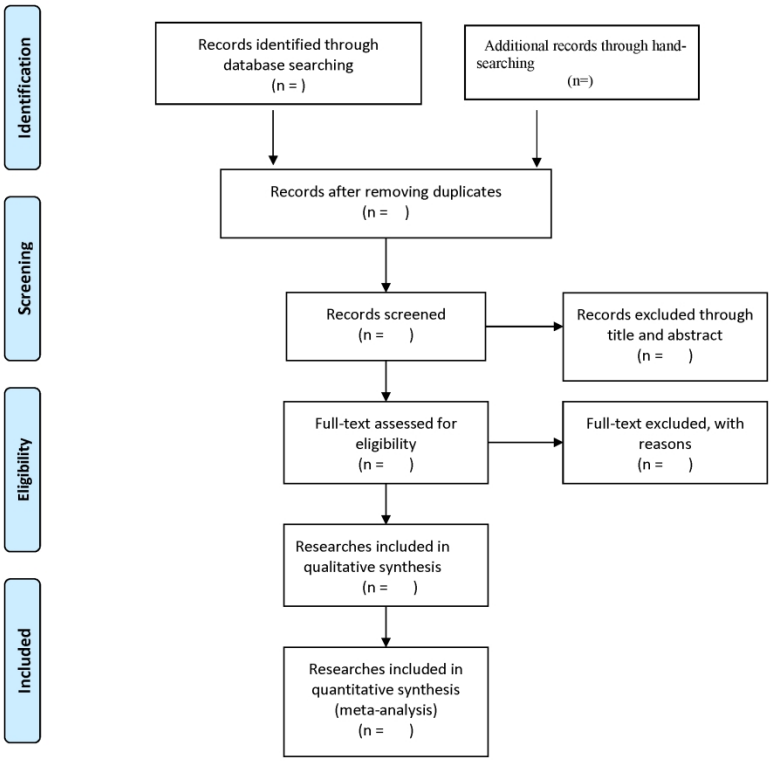


Fig.1 The flow diagram of literature search and selection

215x279mm (300 x 300 DPI)

PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol*

Section and topic	Item No	Checklist item
ADMINISTRATIVE INFORMATION		
Title:		
Identification	1a	Identify the report as a protocol of a systematic review (yes, P1)
Update	1b	If the protocol is for an update of a previous systematic review, identify as such (no)
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number (yes, CRD42021257148, P3, P4)
Authors:		
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author (yes, P1)
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review (yes, P11)
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments (yes, P3)
Support:		
Sources	5a	Indicate sources of financial or other support for the review (yes, P11)
Sponsor	5b	Provide name for the review funder and/or sponsor (yes, P11)
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol (yes, P11)
INTRODUCTION		
Rationale	6	Describe the rationale for the review in the context of what is already known (yes, P3, P4)
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO) (yes, P4)
METHODS		
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review (yes, P7)
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage (yes, P5)
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated (yes, Table1)
Study records:		

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Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review (yes, P8)
Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis) (yes, P8)
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators (yes, P8)
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications (yes, P8)
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale (yes, P7)
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis (yes, P8)
Data synthesis	15a	Describe criteria under which study data will be quantitatively synthesised (yes, P9)
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I^2, Kendall's τ) (yes, P9)
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression) (yes, P9)
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned (yes, P9)
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies) (yes, P9)
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE) (yes, P9)

*** It is strongly recommended that this checklist be read in conjunction with the PRISMA-P Explanation and Elaboration (cite when available) for important clarification on the items. Amendments to a review protocol should be tracked and dated. The copyright for PRISMA-P (including checklist) is held by the PRISMA-P Group and is distributed under a Creative Commons Attribution Licence 4.0.**

From: Shamseer L, Moher D, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart L, PRISMA-P Group. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation. BMJ. 2015 Jan 2;349(jan02 1):g7647.

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